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EXPERIMENT 4

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Semester: 5
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Subject Code: 24CSH-337

Experiment

Implementation of Static Routing across three interconnected routers in Cisco Packet Tracer, so that hosts on six separate LANs can reach one another.

Aim

To configure Static Routing on multiple routers in Cisco Packet Tracer and verify that hosts on different networks can communicate with each other.

Objective

- To understand what routing is and why it is required between separate networks.
- To study how static routing works on a Cisco router.
- To manually configure static routes using the ip route command.
- To establish end-to-end communication across multiple LANs connected through routers.
- To verify connectivity using the Ping command.

Software Requirements

Software

- Cisco Packet Tracer

Hardware (Virtual)

- Routers (Cisco 2911)
- Switches (2960-24TT)
- PCs
- Copper Straight-Through Cables
- Serial/WAN links between routers

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Experiment Theory

Routing is the process by which a router selects a path and forwards packets from a source network toward a destination network that it is not directly connected to. In Static Routing, the administrator manually enters each route into the router's routing table using the ip route command, specifying the destination network, its subnet mask, and the next-hop address. Static routes are simple to configure, do not consume CPU or bandwidth on route computation, and are well suited to small, stable topologies, but unlike dynamic routing protocols they do not automatically adjust if the topology changes.

Practical / Experiment Steps

1. Open Cisco Packet Tracer.
2. Place three routers, three switches, and six PCs on the workspace.
3. Connect each router to its own switch, and connect two PCs to each switch.
4. Link the three routers together in a chain using serial/WAN connections.
5. Configure IP addresses on all PC network adapters and on every router interface.
6. Configure a static route on each router, using the ip route command, for every remote network it is not directly connected to.
7. Save the router configuration.
8. Test communication between PCs on different LANs using the Ping command.
9. Verify that packet transmission across the routers is successful.

Procedure

1. Design the required network topology with three routers, three switches, and six PCs.
2. Configure IP addresses on all end devices.
3. Configure all router interfaces (LAN-facing and inter-router links).
4. Add static routes manually on every router for all non-directly-connected networks.
5. Verify each router's routing table.
6. Perform Ping tests between PCs on different networks.
7. Observe and record successful end-to-end communication.

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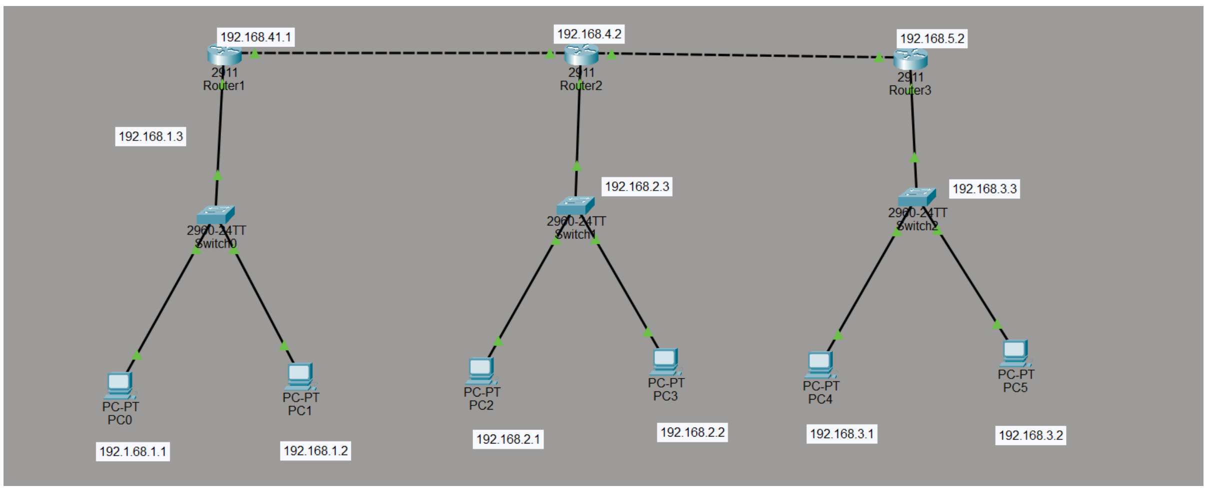
Input / Output Details

Devices Used

- 3 Routers (Cisco 2911)
- 3 Switches (2960-24TT)
- 6 PCs
- Copper Straight-Through Cables
- Serial/WAN Cables between routers

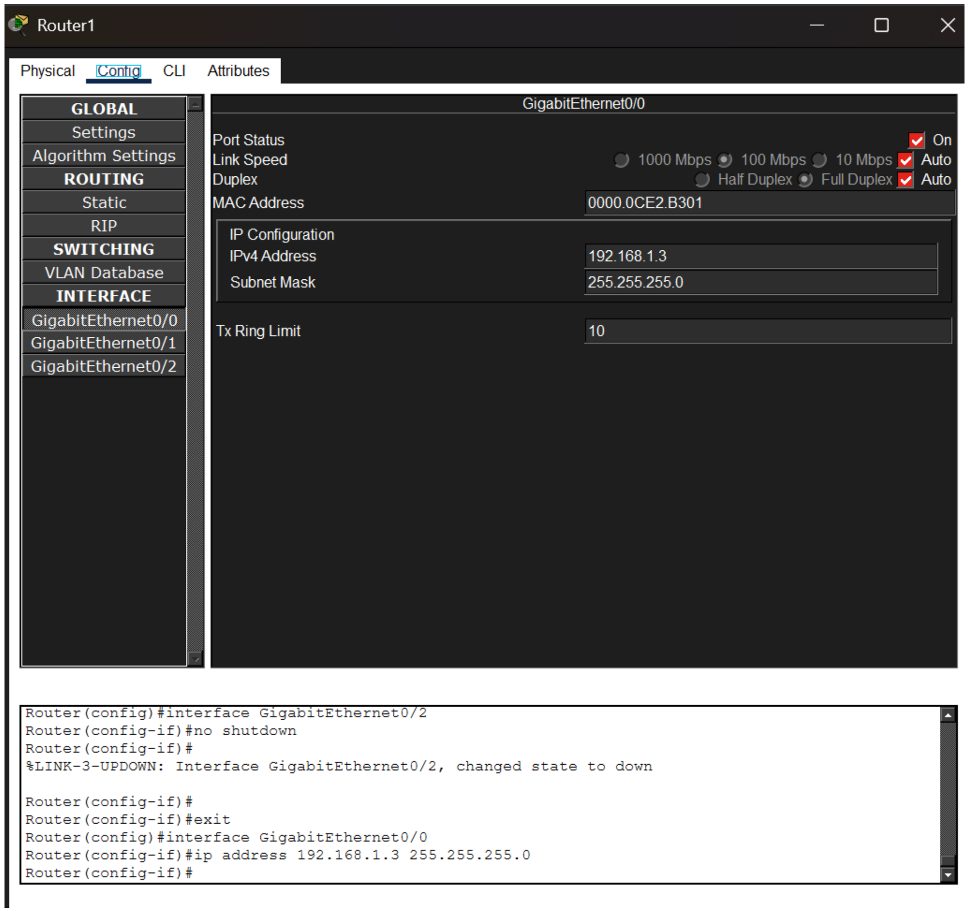
Step-wise Output

Step 1 – Network Topology



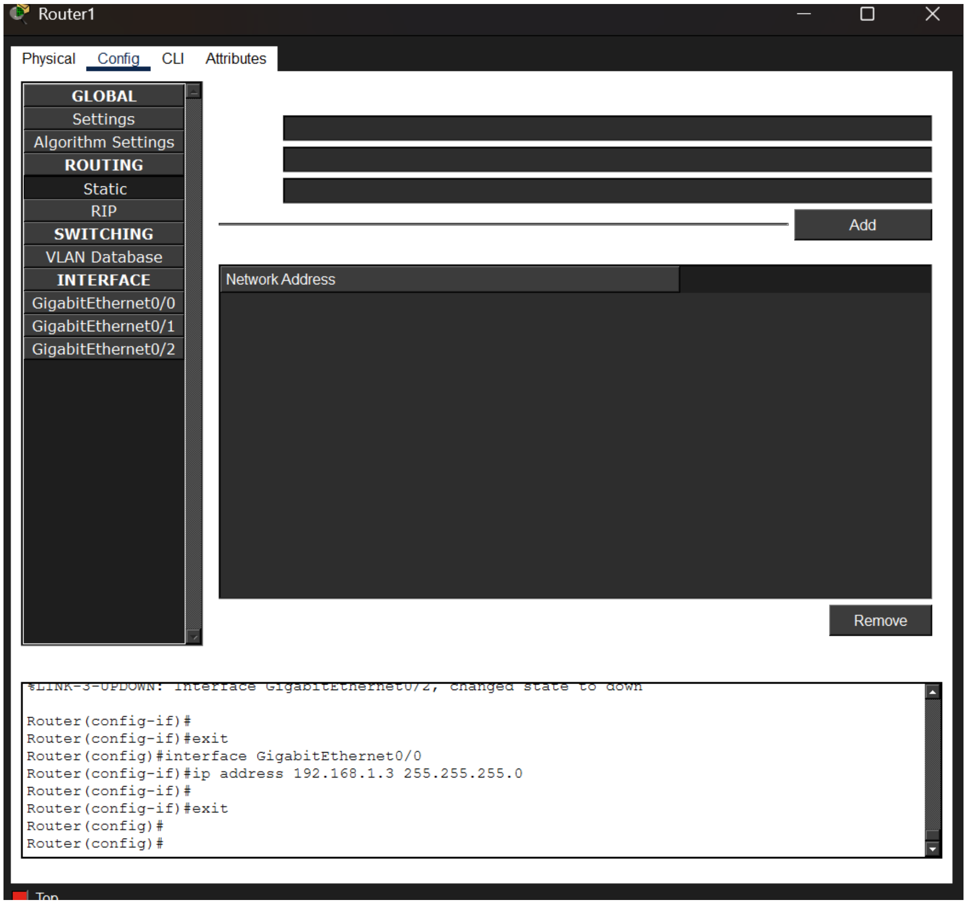
Three routers (Router1, Router2, Router3) are chained together over WAN links: Router1–Router2 on 192.168.4.0/24 and Router2–Router3 on 192.168.5.0/24. Each router also connects to a local switch: Switch0 (192.168.1.0/24) with PC0 and PC1, Switch1 (192.168.2.0/24) with PC2 and PC3, and Switch2 (192.168.3.0/24) with PC4 and PC5.

Step 2 – IP Address Configuration on Router Interfaces



Each router's LAN-facing interface (GigabitEthernet0/0) is assigned an address on its local subnet — here Router1's interface is set to 192.168.1.3 / 255.255.255.0 — while the CLI pane confirms the ip address command being applied directly through the router's configuration terminal.

Step 3 – Static Route Configuration



On the Routing > Static page, a route is added for every remote network the router is not directly connected to, giving the destination network, subnet mask, and the next-hop IP address of the neighboring router. The CLI history confirms these entries were pushed with the ip route command before moving on to configure the next interface.

Learning Outcome

After performing this experiment, I was able to:

- Understand the concept of routing and static routing.
- Configure IP addresses on routers and PCs.
- Configure static routes manually using Cisco IOS commands.
- Verify routing tables and confirm network connectivity.
- Successfully establish communication across multiple networks using static routing.